

Supplementary Material: What the Error Is and Who Makes It: Auditable Misconception Analysis from Distractor Responses

Abstract

This supplementary material provides robustness checks and supporting analyses for the main paper, including encoder comparisons, external misconception validation, behavioural analyses, interface details, and held-out assignment reliability.

Notes for Practice

Keywords

learning analytics, distractor analysis, misconception analysis, semantic clustering, behavioural analysis

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1. Semantic Encoder Robustness

Table 1 compares the semantic encoders at a common $k = 8$, allowing the recovered structures to be evaluated at the same granularity. Qwen3-Embedding-8B gives the highest topic NMI and external misconception-family alignment while showing the weakest association with text length among the tested encoders. Its bootstrap stability is 0.718.

Table 1. Semantic-encoder comparison at fixed $k = 8$. External alignment is reported using MiniLM and a Qwen3-based matcher. Core-15 alignment is calculated from the 15 distractors nearest to each semantic-group centroid.

Encoder	Dim.	Bootstrap ARI	Topic NMI	ARI with behavioural	MiniLM align.	Qwen3 align.	Core-15 align.	η^2 length
Qwen3-Embedding-8B	4096	0.718	0.577	0.060	83.8%	87.7%	89.2%	0.12
bge-large-en-v1.5	1024	0.658	0.387	0.027	82.3%	81.4%	86.7%	0.57
all-mpnet-base-v2	768	0.728	0.436	0.052	81.4%	83.3%	81.7%	0.42
mxbai-embed-large-v1	1024	0.694	0.386	0.028	79.9%	79.3%	83.3%	0.55
MiniLM-L6-v2	384	0.751	0.446	0.038	79.5%	82.0%	84.2%	0.40
Qwen3-Embedding-0.6B	1024	0.699	0.480	0.060	77.1%	78.3%	85.8%	0.37
e5-large-v2	1024	0.605	0.353	0.023	76.8%	77.6%	80.0%	0.51
Qwen3-Embedding-4B	2560	0.617	0.442	0.061	76.6%	79.1%	84.2%	0.38
nomic-embed-text-v1.5	768	0.699	0.344	0.011	76.3%	76.4%	85.0%	0.48
SciBERT	768	0.635	0.249	0.019	65.7%	66.0%	73.3%	0.60

Several encoders achieve their highest bootstrap stability at coarser values of k , particularly $k = 4$. We use the common $k = 8$ setting to preserve the granularity required for candidate-level analysis while keeping the encoder comparison consistent.

2. Combined Semantic and Behavioural Representations

We also examine direct combinations of the semantic and behavioural feature sets by varying their relative contribution. The weighting analysis tests whether an intermediate representation produces a stable grouping that is distinct from both the semantic and behavioural endpoints.

As the behavioural contribution increases, similarity to the semantic grouping decreases while similarity to the behavioural grouping increases. Intermediate weighting settings do not consistently produce a stable grouping that is distinct from both endpoints. Stability also does not show a clear intermediate optimum corresponding to a separate joint structure.

The original combined configuration with $\beta = 10$ is strongly behaviourally weighted, with the behavioural feature block accounting for 97.1% of total feature variance. These results provide little evidence that direct feature combination recovers a separate and reproducible semantic-behavioural grouping. The main workflow therefore retains semantic candidate structure and learner-response behaviour as separate sources of information.

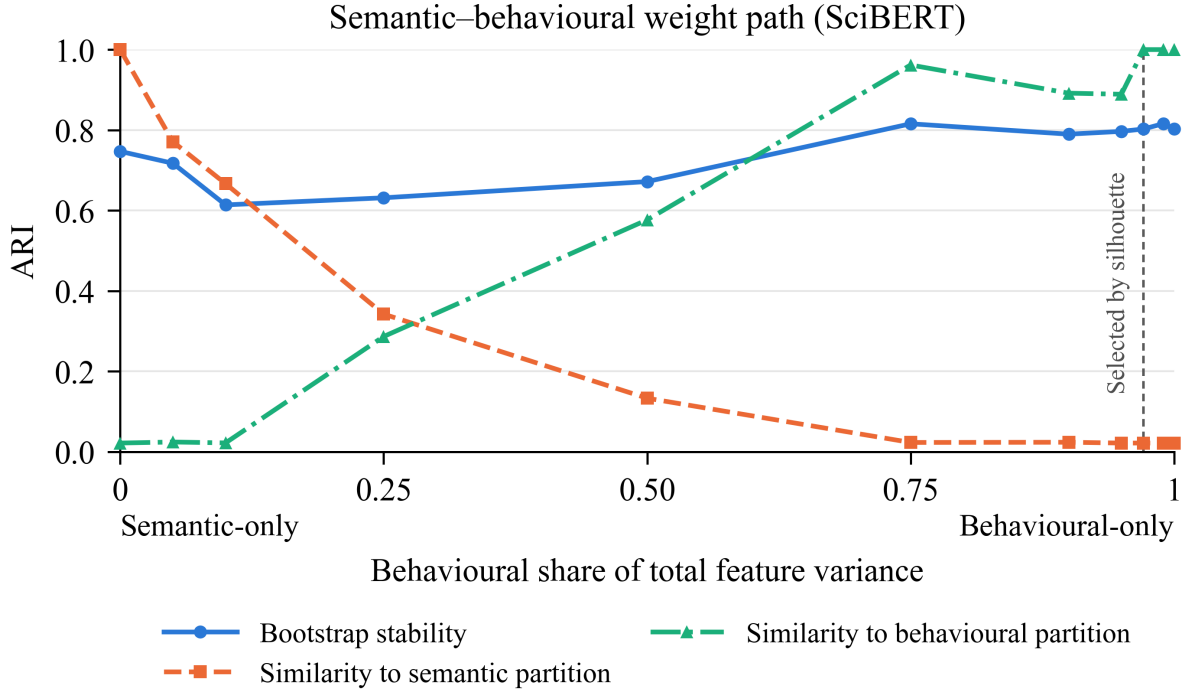


Figure 1. Semantic-behavioural weighting path for the exploratory SciBERT combination analysis. The horizontal axis shows the behavioural share of total feature variance. Bootstrap stability is compared with similarity to the semantic and behavioural endpoint groupings. Increasing the behavioural contribution moves the recovered grouping from the semantic endpoint towards the behavioural endpoint.

3. External Misconception Evidence

S3.1 Does Semantic Grouping Capture More Than Mathematical Topic?

Semantic groups may align with the external misconception bank simply because they contain distractors from the same mathematical topics. We therefore compare the semantic grouping with groups formed directly from the existing Eedi topic labels.

For the main eight-group comparison, the seven most frequent level-2 Eedi topics are retained as separate groups and the remaining topics are combined into an eighth group. We also use a within-topic permutation baseline, in which semantic-group assignments are shuffled only among distractors from the same Eedi topic. This keeps the topic composition unchanged while removing the original semantic grouping within each topic. A random-grouping baseline is included for comparison.

To check whether the result depends on the number of groups, we repeat the comparison using 25 and 56 topic groups. We also calculate external misconception-family alignment using both MiniLM and Qwen3-Embedding-0.6B as misconception matchers.

Table 2. External misconception-family alignment for Eedi topic groupings and Qwen3-Embedding-8B semantic groupings. The within-topic baseline preserves mathematical topic while shuffling semantic-group assignments within each topic.

Groups	Matcher	Eedi topic grouping	Qwen3-8B grouping	Within-topic baseline	Random grouping
8	MiniLM	79.5%	83.8%	79.5%	53.2%
8	Qwen3-Embedding-0.6B	84.5%	87.7%	82.2%	53.4%
25	MiniLM	85.0%	85.2%	81.0%	57.1%
25	Qwen3-Embedding-0.6B	84.1%	87.6%	82.9%	56.8%
56	MiniLM	86.7%	89.0%	81.6%	60.7%
56	Qwen3-Embedding-0.6B	87.1%	87.4%	83.5%	60.2%

At eight groups, the semantic grouping reaches 83.8% external misconception-family alignment with the MiniLM matcher, compared with 79.5% for both the Eedi topic grouping and the within-topic permutation baseline. The semantic grouping is also higher than the within-topic baseline when Qwen3-Embedding-0.6B is used as the matcher. The same pattern is generally

retained at 25 and 56 groups. Across both matchers, the semantic grouping remains above the corresponding within-topic baseline.

These results show that mathematical topic explains a substantial part of the external misconception-family alignment. The semantic grouping nevertheless retains information beyond topic alone, because its alignment remains higher after comparison with groups that preserve the same topic structure.

4. Additional Behavioural Analyses

S4.1 Learner-Pattern Labels

The educator-facing cards use short descriptive labels to summarise three behavioural dimensions analysed in the main paper: the relationship between distractor selection and learner proficiency, learner-specific recurrence across quizzes, and confidence in incorrect responses. These labels summarise the reported statistics and do not introduce additional statistical tests.

For proficiency, an odds ratio (OR) of at most 0.60 is described as a steep decline, values up to 0.66 as a decline, and larger values as a more gradual decline. Groups for which at least 16% of the strongest proficiency quartile still select the corresponding distractors are described as remaining comparatively common among stronger learners.

Learner-specific recurrence is described as clear when the 95% confidence interval excludes zero for both all attempts and incorrect responses only. It is described as partial when this occurs for all attempts but not for incorrect responses only. Groups without either pattern are described as showing little evidence of learner-specific recurrence.

For confidence, a distractor is treated as a confident error when the mean confidence of learners selecting it is within five points of the mean confidence of correct responders. Group-level proportions of at least 30% are described as relatively high, values from 18% to 30% as occasional, and lower values as usually low. No group-level confidence description is assigned when fewer than 30% of the group's distractors have sufficient ratings. Table 3 reports the resulting learner-pattern summaries for all eight semantic candidate groups.

Table 3. Learner-pattern summaries across the eight semantic candidate groups. The labels combine proficiency association, learner-specific recurrence, and confidence in incorrect responses.

Group	Candidate error	Learner pattern	Behavioural basis
S0	Negative-number sign errors	Learner-specific recurring error	Selection declines from 26% to 12% across the weakest and strongest proficiency quartiles (OR = 0.66). Recurrence is present for all attempts (+0.08) and incorrect responses only (+0.07). Confident errors account for 17%.
S1	Fraction representation	Strongly proficiency-linked, weakly recurrent	Selection declines from 30% to 12% (OR = 0.57), with little learner-specific recurrence (+0.01 / +0.01). Too few distractors have sufficient confidence ratings for a group-level confidence summary.
S2	Mixed shapes and symmetry	Broad-proficiency, high-confidence mixed pattern	Selection remains comparatively common among stronger learners (27% to 17%; OR = 0.68), with little learner-specific recurrence (+0.00 / +0.02) and a high proportion of confident errors (44%). The candidate group is not coherent.
S3	Mixed arithmetic	Proficiency-linked, with limited error-specific recurrence	Selection declines from 27% to 12% (OR = 0.63). Recurrence is present across all attempts (+0.06) but not when restricted to incorrect responses (+0.01). Confident errors account for 15%. The candidate group is not coherent.
S4	Area and perimeter	Broad-proficiency, relatively high-confidence pattern	Selection remains comparatively common among stronger learners (29% to 18%; OR = 0.66), with little learner-specific recurrence (+0.00 / +0.01) and a relatively high proportion of confident errors (40%).
S5	Coordinates and transformations	Moderately learner-specific pattern	Selection declines from 27% to 15% (OR = 0.67). Recurrence is present across all attempts (+0.08) but weaker for incorrect responses only (+0.03). Confident errors account for 32%.
S6	Angle size from diagram	Recurring pattern, weaker error-specific recurrence	Selection declines from 26% to 13% (OR = 0.65). Recurrence is present across all attempts (+0.10) but weaker for incorrect responses only (+0.04). Confident errors account for 21%.
S7	Factors vs. multiples	Proficiency-sensitive with learner-specific recurrence	Selection declines from 30% to 11% (OR = 0.59). Recurrence is present for both all attempts (+0.11) and incorrect responses only (+0.07). Confident errors account for 15%.

S4.2 Proficiency Trends

Figure 2 shows how distractor selection changes across learner-proficiency deciles. Panel (a) considers all attempts and therefore shows the overall prevalence of each candidate error pattern across proficiency. Panel (b) considers only incorrect responses and shows how the distribution of wrong answers changes with proficiency. Panel (c) reports the corresponding distractor-level odds ratios per one-standard-deviation increase in proficiency.

Across all groups, focal-distractor selection decreases at higher proficiency when all responses are considered. The decline is most visible among the highest-proficiency learners. Among incorrect responses, however, focal-distractor selection generally increases with proficiency, indicating that higher-proficiency learners who answer incorrectly are more likely to concentrate on particular distractors instead of distributing their errors across the available wrong options.

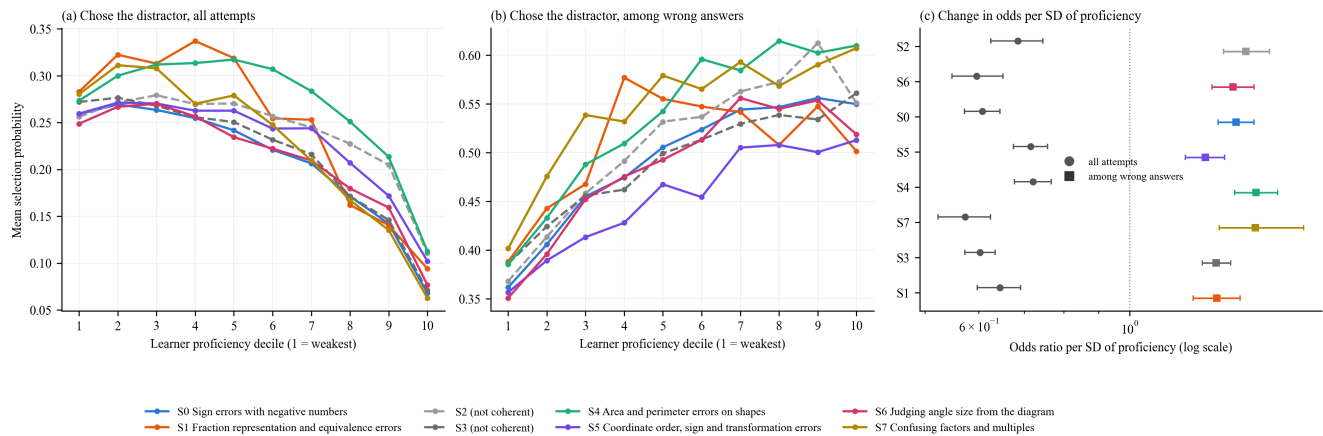


Figure 2. Distractor selection across learner proficiency. Panel (a) shows mean selection probability across all attempts, Panel (b) shows focal-distractor selection among incorrect responses only, and Panel (c) reports the corresponding odds ratios. Values below one indicate decreasing selection with proficiency; values above one in the incorrect-response analysis indicate greater concentration on the focal distractor among higher-proficiency learners who answer incorrectly.

S4.3 Learner-Specific Recurrence

Figure 3 provides additional detail on learner-specific recurrence. Recurrence is calculated from proficiency-adjusted selection residuals across separate sets of quizzes, so that the measure reflects repeated group-specific errors beyond general differences in proficiency.

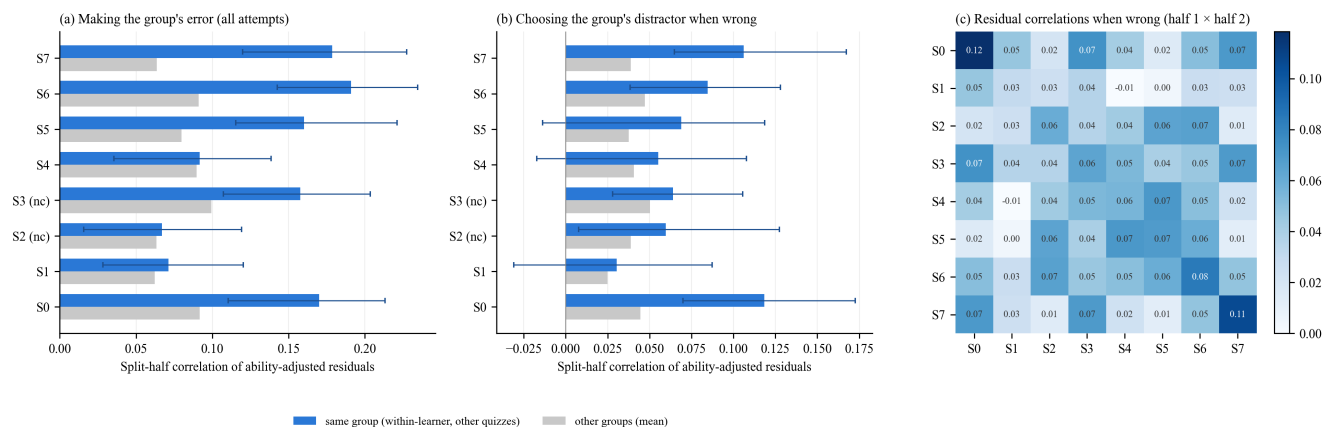


Figure 3. Learner-specific recurrence across separate sets of quizzes. Panel (a) shows within-group split-half correlations and the mean correlation with other groups using all attempts. Panel (b) repeats the analysis using incorrect responses only. Panel (c) shows cross-group residual correlations for the incorrect-response analysis. Higher within-group correlations indicate stronger learner-specific recurrence.

Negative-number sign errors (S0) and factor–multiple confusion (S7) show the clearest learner-specific recurrence, including when the analysis is restricted to incorrect responses. Coordinate and transformation errors (S5) and angle-size errors (S6) show recurrence across all attempts, with weaker evidence for incorrect responses only. Fraction representation errors (S1) and area and perimeter errors (S4) show little learner-specific recurrence.

These results show that proficiency and learner-specific recurrence provide different information about how candidate errors appear across learners.

5. Educator-Facing Interface

The educator-facing interface supports four related tasks: reviewing an individual candidate error group, comparing learner-response patterns across groups, inspecting the external evidence used in the candidate audit, and rating candidate cards during the human evaluation. The interface keeps generated interpretations, learner-response evidence, and external audit evidence visibly separated.

S5.1 Candidate-Group Review

Figure 4 shows the main review workspace for a supported candidate error group. The page brings together the generated interpretation, evidence-linked claims, representative distractors, and the learner-pattern summary. The audit status is shown alongside the candidate interpretation, while the detailed audit evidence remains in a separate expandable panel.

S5.2 Cross-Group Comparison

Figure 5 shows the cross-group comparison view. Each candidate group is positioned according to its difficulty-adjusted proficiency sensitivity and learner-specific recurrence. Point size represents the proportion of confident errors, and point shape indicates candidate-audit status. The accompanying table reports the corresponding group-level statistics.

S5.3 External Audit Evidence

Figure 6 shows the external-evidence panel used to inspect the support for a candidate group. The panel reports the dominant external misconception family, full-group and core-15 concentration, permutation-based comparisons with random groups, and the most frequently retrieved misconception descriptions. These external matches are used for candidate audit and are not included in the evidence supplied to the LLM interpretation.

S5.4 Human-Evaluation Interface

Figure 7 shows the interface used during the human evaluation. Raters viewed the candidate interpretation, evidence-linked claims, representative distractors, and the behavioural summary before assigning scores for Accuracy, Specificity, Completeness, Evidence Grounding, and Pedagogical Usefulness. The interface also provided an optional comment field.

Candidate Error Review Workspace

Candidate error patterns in multiple-choice distractors: descriptive evidence, not diagnoses of individual learners

Review

Compare groups

Evaluation mode

CANDIDATE GROUPS

- S0 Negative-number signs**
 ✓ SUPPORTED
 proficiency: moderate decline · recurrence: clear
- S1 Fraction representation**
 ! PARTIALLY SUPPORTED
 proficiency: strong decline · recurrence: little
- S2 Mixed shapes/symmetry**
 - NOT COHERENT
 proficiency: weaker decline · recurrence: little
- S3 Mixed arithmetic**
 - NOT COHERENT
 proficiency: moderate decline · recurrence: some
- S4 Area and perimeter**
 ✓ SUPPORTED
 proficiency: moderate decline · recurrence: little
- S5 Coordinates/transformation**
 ✓ SUPPORTED
 proficiency: weaker decline · recurrence: clear
- S6 Angle estimation**
 ✓ SUPPORTED
 proficiency: moderate decline · recurrence: clear
- S7 Factors vs multiples**
 ✓ SUPPORTED
 proficiency: strong decline · recurrence: clear

CANDIDATE ERROR GROUP S7

Confusion between factors and multiples

✓ SUPPORTED Why? +

Learner pattern: Proficiency-sensitive with learner-specific recurrence

91 distractors Bootstrap stability 0.87 External evidence: strong ✓ Evidence-link audit passed

GENERATED INTERPRETATION

Candidate interpretation written by an LLM from the representative distractors and response summaries only.

The distractors are statements about factors, common factors and multiples. The chosen answers reject correct factor statements or accept incorrect ones, consistent with confusing factors and multiples.

EVIDENCE-LINKED CLAIMS

- Correct statements that 1 is a factor of a number are rejected, for 12 and for 9. [e4](#) [e5](#)
- Factors of a number written in prime-factor form are misjudged, such as rejecting that 6 is a factor of $2^2 \times 3^3 \times 5^2$ or accepting that 8 is. [e1](#) [e9](#)
- When asked which number is not a factor, an actual factor is chosen (4 for 12, 9 for 36). [e14](#) [e11](#)
- When asked which number has 3, 9 and 18 as factors, 3 is chosen instead of 18, reversing the relation between a number and its factors. [e15](#)
- Statements about common factors and common multiples are also misjudged. [e3](#) [e6](#) [e13](#)

Evidence-link audit: 5 substantive claims, 5 linked to valid exemplar IDs, 0 unresolved references. This checks traceability to the supplied examples, not the correctness of the interpretation.

EVIDENCE: REPRESENTATIVE DISTRACTORS

Question + correct answer + selected distractor, from the 15 distractors nearest the group centre.

Example e4 cited in claim 1 Factors, Multiples and Primes
 Tom and Katie are arguing about the factors of 12. Tom says 1 is a factor of 12. Katie says 12 is a factor of 12. Who is correct?

Correct answer ✓ Both Tom and Katie

Selected distractor ✗ Only Katie

selected on 7% of 2,842 attempts 17% answered incorrectly

[Show question image](#)

Example e5 cited in claim 1 Factors, Multiples and Primes
 Tom and Katie are arguing about the factors of 9. Tom says 1 is a factor of 9. Katie says 9 is a factor of 9. Who is correct?

Correct answer ✓ Both Tom and Katie

Selected distractor ✗ Only Katie

selected on 12% of 1,734 attempts 30% answered incorrectly

[Show question image](#)

Example e1 cited in claim 2 Factors, Multiples and Primes
 Jo and Paul are still arguing about the number expressed as $2^2 \times 3^3 \times 5^2$. Jo says 6 is a factor of the number. Paul says 45 is a factor of the number. Who is correct?

Correct answer ✓ Both Jo and Paul

Selected distractor ✗ Only Paul

selected on 29% of 85 attempts 67% answered incorrectly

[Show question image](#)

Example e9 cited in claim 2 Factors, Multiples and Primes
 Jo and Paul are arguing about a number which has been expressed in the following way: $2^2 \times 3^3 \times 5^2$. Jo says 8 is a factor of the number. Paul says 3 is a factor of the number. Who is correct?

Correct answer ✓ Only Paul

Selected distractor ✗ Both Jo and Paul

selected on 24% of 85 attempts 49% answered incorrectly

[Show question image](#)

Example e14 cited in claim 3 Factors, Multiples and Primes
 Which of these is not a factor of 12?

Correct answer ✓ Twenty-four

Selected distractor ✗ Four

selected on 9% of 2,325 attempts 22% answered incorrectly

[Show question image](#)[Show all 15 representative examples](#)

HOW THIS ERROR APPEARS ACROSS LEARNERS Patterns across the learner population, not profiles of individual learners

HOW DOES THIS CHANGE WITH PROFICIENCY? ?

Steep decline with proficiency

30% → 11%

One of the strongest declines across candidate groups.

DOES IT RECUR IN THE SAME LEARNERS? ?

Yes — recurs across quizzes

+0.11 overall

+0.07 among incorrect responses

The same learners tend to show this pattern across separate quizzes.

ARE LEARNERS CONFIDENT WHEN WRONG? ?

Usually not

15%

Among the lower confidence rates across candidate groups.

What this pattern may indicate. More concentrated among lower-proficiency learners and shows learner-specific recurrence across quizzes. This pattern may warrant closer inspection for learners who repeatedly select similar distractors.

Observational summaries of response data; they describe groups of responses, not the misconceptions of individual learners, and do not show that any intervention would help.

[► Details and comparison with other groups](#)

► Why is this candidate supported?

candidate audit

► External audit evidence

kept separate from the LLM interpretation

Figure 4. Full educator-facing review workspace for candidate group S7. The interface presents the generated interpretation and evidence-linked claims alongside representative distractors and a summary of how the error appears across learners. Candidate-audit and external-evidence panels are kept separate from the generated interpretation.

Candidate Error Review Workspace

Candidate error patterns in multiple-choice distractors: descriptive evidence, not diagnoses of individual learners

Review

Compare groups

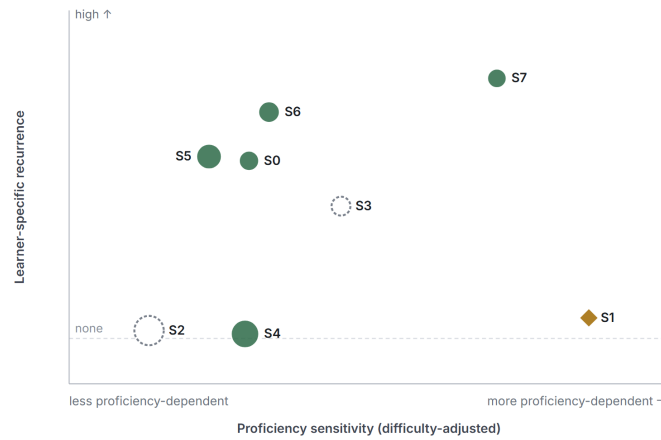
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- S7 Factors vs multiples**
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COMPARE CANDIDATE GROUPS: BEHAVIOURAL LANDSCAPE

Each point is a semantic candidate group. Horizontal: how strongly selection declines with proficiency (1 – adjusted odds ratio per SD). Vertical: recurrence within the same learners across quizzes. Point size: share of confident errors (smallest point = insufficient ratings). Shape: ● supported, ● partially supported, ○ not coherent. Select a point to open the group.



GROUP	STATUS	OR/SD	RECUR.	INCCOR.	CONF.
S0 Negative-number signs	✓ SUPPORTED	0.66	+0.08	+0.07	17%
S1 Fraction representation	! PARTIALLY SUPPORTED	0.57	+0.01	+0.01	–
S2 Mixed shapes/symmetry	– NOT COHERENT	0.68	+0.00	+0.02	44%
S3 Mixed arithmetic	– NOT COHERENT	0.63	+0.06	+0.01	14%
S4 Area and perimeter	✓ SUPPORTED	0.66	+0.00	+0.01	40%
S5 Coordinates/transformation	✓ SUPPORTED	0.67	+0.08	+0.03	32%
S6 Angle estimation	✓ SUPPORTED	0.65	+0.10	+0.04	21%
S7 Factors vs multiples	✓ SUPPORTED	0.59	+0.11	+0.07	15%

The two dimensions describe how the distractors are selected across learners. They do not diagnose misconceptions: a group can be proficiency-linked without recurring in the same learners, and the reverse.

Figure 5. Cross-group behavioural comparison in the educator-facing interface. The horizontal axis represents difficulty-adjusted proficiency sensitivity and the vertical axis represents learner-specific recurrence across quizzes. Point size reflects the share of confident errors, and point shape indicates candidate-audit status. The table below the plot reports the corresponding group-level behavioural measures.

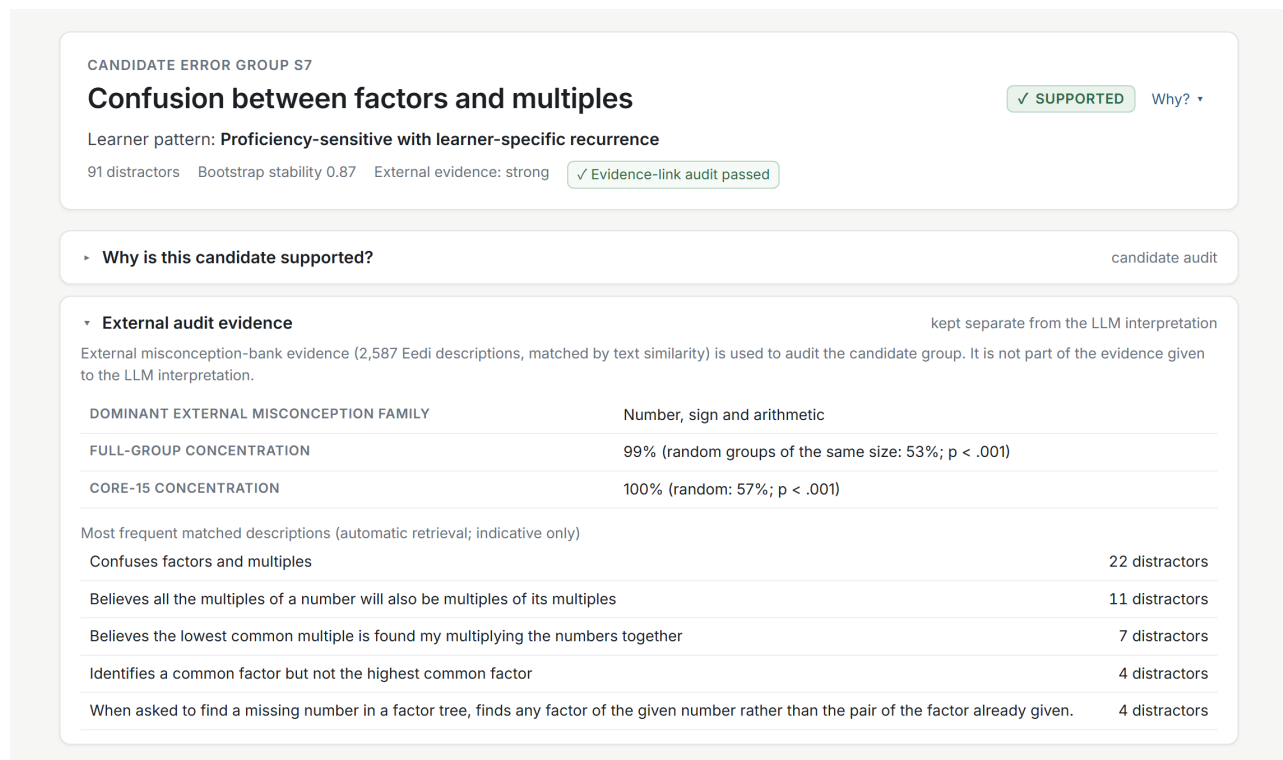


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Candidate error patterns in multiple-choice distractors: descriptive evidence, not diagnoses of individual learners

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Rater ID

Condition

Evidence-linked

Card 8 of 8

Export JSON

Export CSV

CANDIDATE CARD S7

THEME

Confusion between factors and multiples

SUMMARY

The distractors are statements about factors, common factors and multiples. The chosen answers reject correct factor statements or accept incorrect ones, consistent with confusing factors and multiples.

CLAIMS

- Correct statements that 1 is a factor of a number are rejected, for 12 and for 9. [e4](#) [e5](#)
- Factors of a number written in prime-factor form are misjudged, such as rejecting that 6 is a factor of $2^2 \times 3^3 \times 5^2$ or accepting that 8 is. [e1](#) [e9](#)
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- Statements about common factors and common multiples are also misjudged. [e3](#) [e6](#) [e13](#)

REPRESENTATIVE EXAMPLES

Example e4

Factors, Multiples and Primes

Tom and Katie are arguing about the factors of 12. Tom says 1 is a factor of 12. Katie says 12 is a factor of 12. Who is correct?

Correct answer ✓ Both Tom and Katie

Selected distractor ✗ Only Katie

selected on 7% of 2,842 attempts 17% answered incorrectly [Show question image](#)

Example e1

Factors, Multiples and Primes

Jo and Paul are still arguing about the number expressed as $2^2 \times 3^3 \times 5^2$. Jo says 6 is a factor of the number. Paul says 45 is a factor of the number. Who is correct?

Correct answer ✓ Both Jo and Paul

Selected distractor ✗ Only Paul

selected on 29% of 85 attempts 67% answered incorrectly [Show question image](#)

Example e14

Factors, Multiples and Primes

Which of these is not a factor of 12?

Correct answer ✓ Twenty-four

Selected distractor ✗ Four

selected on 9% of 2,325 attempts 22% answered incorrectly [Show question image](#)

Example e5

Factors, Multiples and Primes

Tom and Katie are arguing about the factors of 9. Tom says 1 is a factor of 9. Katie says 9 is a factor of 9. Who is correct?

Correct answer ✓ Both Tom and Katie

Selected distractor ✗ Only Katie

selected on 12% of 1,734 attempts 30% answered incorrectly [Show question image](#)

Example e9

Factors, Multiples and Primes

Jo and Paul are arguing about a number which has been expressed in the following way: $2^2 \times 3^3 \times 5^2$. Jo says 8 is a factor of the number. Paul says 3 is a factor of the number. Who is correct?

Correct answer ✓ Only Paul

Selected distractor ✗ Both Jo and Paul

selected on 24% of 85 attempts 49% answered incorrectly [Show question image](#)

BEHAVIOURAL SUMMARY

Distractors in this group were selected on 24% of attempts on average: 30% of attempts by the weakest quarter of learners and 11% by the strongest.

RATINGS (1 = VERY POOR, 5 = VERY GOOD)

Accuracy	1	2	3	4	5
Specificity	1	2	3	4	5
Completeness	1	2	3	4	5
Evidence grounding	1	2	3	4	5
Pedagogical usefulness	1	2	3	4	5

Comment (optional)

Previous

Save and next

Figure 7. Human-evaluation interface used by the six raters. Each candidate card is shown with its interpretation, evidence-linked claims, representative distractors, and behavioural summary. Raters score Accuracy, Specificity, Completeness, Evidence Grounding, and Pedagogical Usefulness on five-point scales and may provide an optional comment.

6. Held-Out Assignment Reliability

To assess whether the semantic candidate-group structure learned from the training questions can be applied to unseen questions, we assign distractors from the held-out test set to the semantic groups learned from the training data. Each test distractor is encoded using the same semantic representation and assigned to its nearest training centroid. Because the held-out distractors do not have expert-assigned misconception labels, this analysis evaluates assignment reliability and not classification accuracy.

Reliability is assessed using three complementary measures. Bootstrap assignment agreement is the proportion of 50 bootstrap models that support the most common group assignment for each held-out distractor. Centroid margin measures the separation between the nearest and second-nearest training centroids,

$$M_i = \frac{d_{i,(2)} - d_{i,(1)}}{d_{i,(2)} + \epsilon}, \quad (1)$$

where $d_{i,(1)}$ and $d_{i,(2)}$ are the distances to the nearest and second-nearest centroids. Distance-based coverage measures whether a held-out distractor falls within the group-specific 95th-percentile distance range observed in the training data. Together, these measures describe whether an assignment is stable across resampling, separated from alternative groups, and consistent with the range of examples observed during training.

The held-out test set contains 180 distractors from unseen questions. Table 4 summarises the overall assignment reliability.

Table 4. Overall held-out assignment reliability for the Qwen3-Embedding-8B semantic grouping.

Measure	Result
Mean bootstrap assignment agreement	0.86
Agreement ≥ 0.80	74.4%
Median centroid margin	0.114
95th-percentile distance coverage	92.2%

Table 5 reports assignment counts and bootstrap agreement for each semantic candidate group.

Table 5. Held-out assignment counts and bootstrap agreement by semantic candidate group.

Group	Held-out distractors	Agreement
S0	20	0.85
S1	50	0.89
S2	15	0.53
S3	29	0.80
S4	22	0.94
S5	22	0.91
S6	10	0.99
S7	12	0.97

Most held-out distractors can be assigned reproducibly to the learned semantic structure, although reliability varies across candidate groups. S2 has the lowest bootstrap agreement at 0.53, while S6 has the highest at 0.99.